

Vedant Shinde

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EDUCATION

University at Buffalo

MS in Data Science

Coursework: Machine Learning, Statistical Learning, Probability, Data Models and Query Language, Reinforcement Learning

Buffalo, USA

Aug 2025 – Expected Dec 2026

Vasantdada Patil Pratisthan's College of Engineering and Visual Arts

BE in Information Technology

Mumbai, India

Jun 2017 – Jun 2021

EXPERIENCE

Morgan Stanley

Senior Associate - SRE Automation

Mumbai, India

Oct 2021 – Jul 2025

- Engineered anomaly and volume deviation detection models for MQ flows using statistical methods (**z-score, standard normal**), integrated with Prometheus and Grafana, reducing root-cause analysis time by 40% and preventing 10+ potential failures per month.
- Designed and presented a one-stop "**Definition of Done**" dashboard for 15+ OTC and ETD mandates, enabling senior management to track **100,000+ daily EOD transactions**, improving decision-making speed by 40%.
- Applied **machine learning** models (**Isolation Forest, GMM**) for log-stream anomaly detection, improving incident detection accuracy by **25%** and reducing manual triage effort.
- Automated alert-parsing pipeline querying a **data warehouse** with **regex-based asset detection**, auto-generating **Jira tickets** routed to relevant squads, reducing operational TOIL by **30%**.
- Built a real-time **log telemetry streaming pipeline** using **Prometheus Pushgateway** and **PromQL**, enabling traceability across **50+ Java microservices** and reducing mean-time-to-diagnose (MTTD) by **10%**.
- Led & Streamlined critical operational workflows and implemented **unit tests & chaos tests using Gremlin** for DB, Asset Inventory, and Eligibility Engine, reducing manual effort by **10%** and enabling continuous monitoring of critical engine responses.

PROJECTS

Supply Chain Data Pipeline (2026)

- Implemented an end-to-end **data engineering workflow** ingesting **600,000+ supply chain records** into **PostgreSQL**; modeled a normalized **3NF OLTP schema** using **Python, Pandas, and SQL**.
- Engineered **dbt-based OLTP → OLAP transformations** to produce **star schema** fact and dimension tables with automated data-quality tests; deployed a **Streamlit** dashboard on **Render** for supply chain insights.

Reinforcement Learning (2026)

- Stock Trading Environment:** Applied **Q-learning** to 2 years of historical Nvidia stock data (\$100k capital); engineered state and reward functions, optimized hyperparameters, and visualized performance, demonstrating RL for **financial decision-making and risk management**.
- Warehouse Robot Navigation:** Developed a **Gymnasium-compliant** warehouse environment from scratch; deployed **SARSA, Q-learning, and Deep Q-Networks** agents, increasing **episodic rewards by 10%** and ensuring robust policy performance under stochastic conditions.

RLAIF Flight Booking Dialogue Agent (2026)

- Built a 3-component **RLAIF pipeline** on **Vertex AI** fine-tuning **Mistral-7B + LoRA** via **PPO** over **5,000+ AirDialogue training samples**, with a **Gemini** simulator and AI judge providing autonomous 1–10 reward signals, eliminating human labelling.
- Resolved **float16 overflow** and **PPO ratio divergence** through mixed-precision arithmetic, separate actor-critic optimizers, and gradient clipping, enabling stable end-to-end training on **GCP Vertex AI**.

Online Retail Analysis (2025)

- Performed feature engineering (RFM analysis, one-hot encoding) and data preprocessing on **500k+ transactions**, improving model performance and training efficiency.
- Evaluated **Logistic Regression, XGBoost, and Random Forest**; selected **Random Forest** based on highest **F1-score** and superior robustness to class imbalance.
- Deployed the best-performing model via **FastAPI** for real-time inference, containerized with **Docker** and hosted on **Render**, enabling scalable predictions for high-value customer targeting.

Drowsy Driver Detection (Published in IRJET: Vol. 8 2021)

- Designed a lightweight **CNN-based** eye-state classifier achieving **95% accuracy**, revamped for deployment on edge devices for driver fatigue detection.
- Integrated a real-time alert pipeline triggering in **under 150ms**, significantly reducing response latency and enhancing driver safety.

SKILLS

- ML & AI:** Python, PyTorch, Pandas, Numpy, Scikit-Learn, MLflow, MLOps, Neural Networks, Deep Learning, Reinforcement Learning, LLM, Hypothesis/A/B Testing, ETL, Data Analysis, Statistical Analysis, Data Visualization, GCP/Vertex AI, AWS
- Languages & Databases:** SQL, PostgreSQL, MongoDB, SQLite, DB2, Sybase, PromQL, R, databricks
- Tools & Frameworks:** Docker, FastAPI, Flask, Streamlit, Git, Version Control, Jenkins, Advanced Analytics, Grafana, Predictive Modeling, Render Cloud, Data Engineering, Time Series Data, Unstructured Data, Excel, Teamwork, Leadership